

Why Climate Disasters Don't Guarantee Climate Action?

Linking Regional Extreme Weather Events and Opinion Dynamics to Macroeconomic Outcomes in DSK+OD ABM

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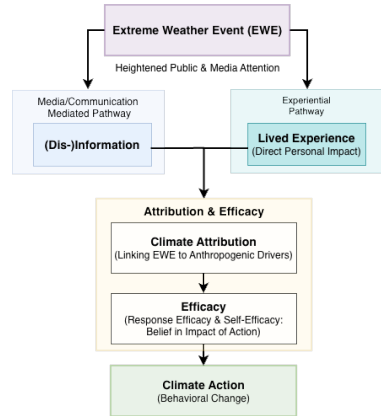
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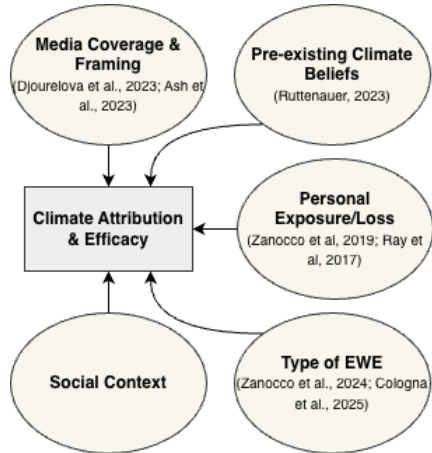


Introduction

- Extreme weather events (EWEs) are increasing in frequency, intensity and economic damages
Intergovernmental Panel On Climate Change (2014); Coronese et al. (2019)
- Public and media attention are heightened during EWE
Roxburgh et al. (2019); Friedman et al. (2019)
- EWEs can serve as catalysts for major societal and political shifts (“critical junction”) Casali et al. (2024)
- Attribution of EWEs to climate change is key for converting experience into mitigation motivation Ogunbode et al. (2019)



What shapes climate attribution post EWE?



Regional heterogeneity in climate impacts & opinions

- Climate damages from EWEs are **uneven across regions**, creating long-run path dependencies in economic performance and climate policy support
Ray et al. (2017); Giordono et al. (2020a)
- Regional EWE exposure differs in frequency as well as type
- National support for climate policy reflects interactions across regions, including:
 - **Opinion dynamics** among affected residents (e.g., climate attribution after shocks)
Cologna et al. (2025)
 - **Economic & demographic weight** of impacted regions (core vs periphery)
Bølstad (2015)
- Heterogeneous exposure & perceptions help explain **variation in local support for climate policies**
Giordono et al. (2020b)

Research Question

How do regionally heterogeneous extreme weather events shape macroeconomic dynamics and climate policy outcomes through opinion dynamics?

- 1 How does heterogeneous regional exposure to extreme weather events affect regional economic trajectories and aggregate national macroeconomic outcomes?

► Phase 1

- 2 How do localized experiences of extreme weather events influence regional and national climate policy preferences, including both the strengthening and erosion of support for mitigation and adaptation policies?

► Phase 2

Model Extension

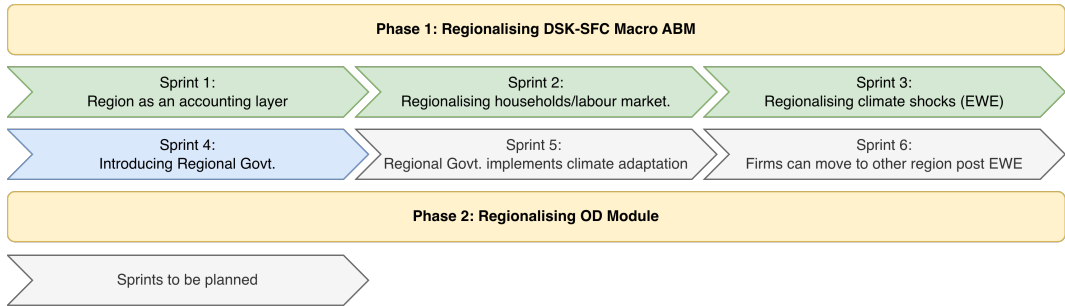
Phase 1: Regionalised Agents and Climate Shocks

- **Objective:** Explore how regional EWE exposures affect national and regional macro-economic outcomes.
- **Modelling:**
 - Regionalised agents: firms, labour (households), and energy plants
 - Capital Market, Goods Market and Energy Market remains at national level (unchanged).
 - Regional governments with budgets and policies.
- **Hypothesis:** Regional EWE damages create persistent economic path dependencies.
- **Policies:**
 - Baseline: no active regional governments.
 - Adaptation: regions offer relief subsidies or invest in damage reduction/rebuilding post EWE.
 - Policy mixes: test interactions of national mitigation and regional adaptation.

Phase 2: Opinion Dynamics and Regional Climate Shocks

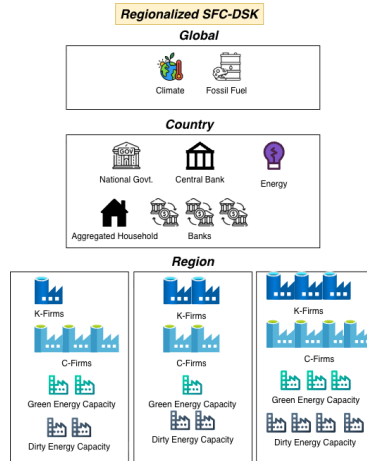
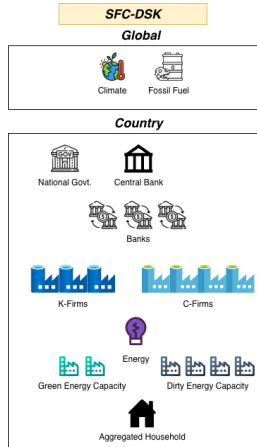
- **Objective:** Endogenize support for climate policies via opinion dynamics.
- **Modelling:**
 - Drivers: EWE exposure, attribution, beliefs, employment, (dis)information.
 - Outcomes: Differentiated support for mitigation (national) vs. adaptation (regional) policies.
- **Hypotheses:**
 - EWE exposure increases adaptation support; mitigation support is conditional
▶ see figure
 - Repeated EWE can either exacerbate or alleviate political polarization and climate fatalism among residents.
- **Policies:**
 - Baseline: Households form opinions on national carbon policy.
 - Adaptation scenario: Opinions include regional adaptation support.
 - Election scenario: Opinion shifts only affect policies during election cycles.

Model Extension Roadmap: Sprint Planning



Regional Firms

Sprint 1: Regionalising K & C Firms and Energy Capacity



Sprint 1: Regionalising K & C Firms and Energy Capacity

- 1 Region is conceptualized as an aggregator/accounting layer in the existing model. It is not an agent and does not influence model behaviour.
- 2 In this first sprint, K-Firms, C-Firms are tagged to a region.
- 3 Energy capacity, production, emissions, etc. are computed at regional level as derivatives based on regional capacity share.
- 4 'AGGREGATE_REGIONS' function is introduced which reports regional aggregates and also act as a verification to ensure regional aggregates sum to global aggregates.
- 5 Other variables and behavioural aspects are left untouched.

Sprint 1: Regionalising K & C Firms

- If "regions" entry is present in the input file, firms are assigned a regional tag at initialization (sequential attribution).
- Upon exit and entry, firms inherit the regional tag associated with the slot they occupy.
- **Assumption** the proportion of K & C-firms does not evolve endogenously over the time.
- **Assumption** The market remains at national level. **No** regional dynamics here!

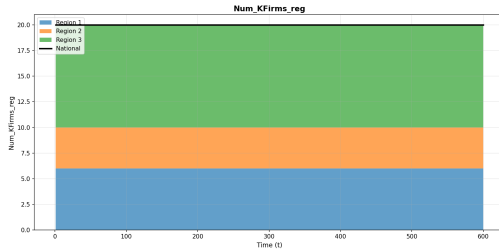


Figure: Number of K-Firms

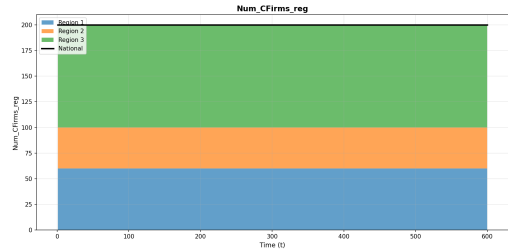


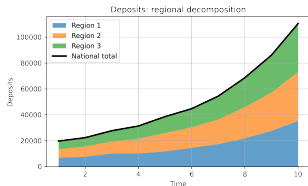
Figure: Number of C-Firms

Sprint 1: Regionalising Energy Capacity (Plants)

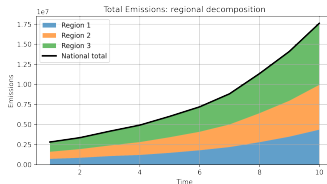
- Regional accounting layer computes following energy sector parameters (derived) at regional level using `dirty_capacity_share` and `green_capacity_share`.
- The energy market is also at national level analogous to a national electricity grid.
- The climate shock to energy capacity (plants) can be implemented using `dirty_capacity_share` and `green_capacity_share` when a region is hit by an EWE/Shock.

Sprint 1: Verifying Regional Aggregates (Sanity Check)

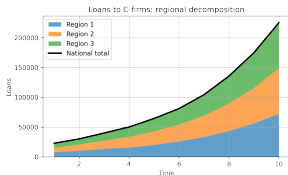
A 10-step simulation with three regions is executed to ensure that regional aggregates align with national-level aggregates.



Deposits



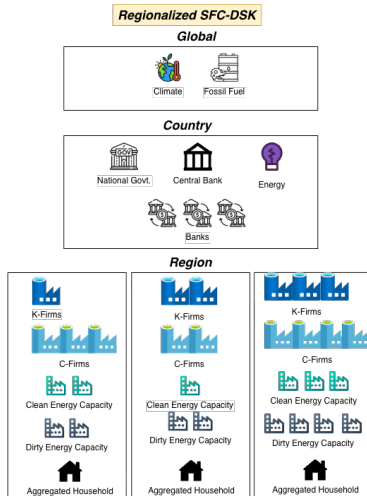
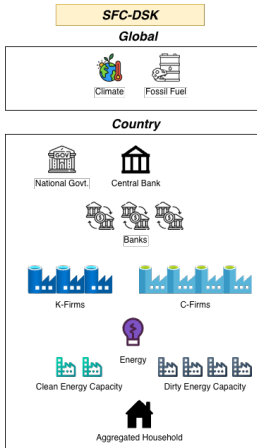
Emissions



loans

Regional Labour

Sprint 2: Regional Labour and Refining Regionalisation of Energy Capacity, K & C Firms



Sprint 2: Refining regional allocation of firms and energy capacity

- Introduced proportion-based regional allocation for:
 - 1 K-firms (static)
 - 2 C-firms (static)
 - 3 Dirty energy capacity & expansion
 - 4 Clean energy capacity & expansion
- Endogenous regional distribution of Labour 'LS0'
- Regional computation of Emissions from K-Firms, C-Firms & Energy sector

Excerpt from 'regions_input.json'

```

...
"regions":
{
  "NR": 3,
  "K_firm_shares": [0.3, 0.2, 0.5],
  "C_firm_shares": [0.3, 0.2, 0.5],

  "energy": {
    "dirty_capacity_shares": [0.1, 0.2, 0.7],
    "green_capacity_shares": [0.5, 0.3, 0.2]}},

  "de_growth_probability": [0.1, 0.2, 0.7],
  "ge_growth_probability": [0.5, 0.3, 0.2]
}
...

```

Sprint 2: Probability based regional allocation for new energy capacity

- Distribution of clean and dirty energy capacity is specified in the 'regions_input.json'
- New capacity expansion is determined based on regional growth probability specified in 'regions_input.json'.

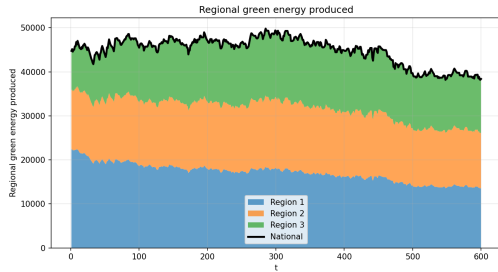


Figure: Green Energy Production

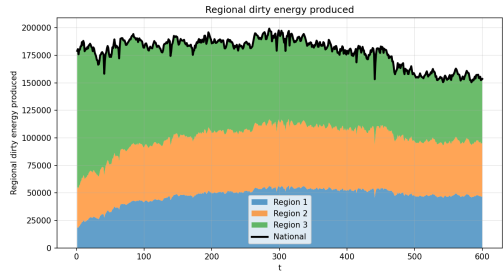


Figure: Dirty Energy Production

Sprint 2: Regionalising Households/Labour

- 1 Regional labour demand (C-Firms, K-Firms and Energy Sector) is computed during initialisation and Labour 'LS0' is distributed across regions in a way that all labour demands are met at 1.
- 2 Regional labour shares are distributed endogenously based on labour demand by K and C firms in that region. This keeps the existing SFC-DSK mechanisms intact.
- 3 **Assumption Free inter-region mobility** If a labour loses employment, it will move to another region with labour demand.
- 4 Labour stays unemployed in a region, until a new labour demand emerges in the same or different region.

Sprint 2: Regionalising Households/Labour

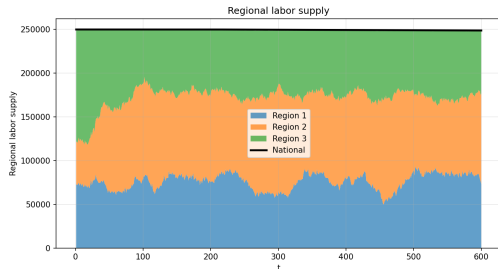


Figure: Labour Supply Pool LS0



Figure: Employment Rate

Sprint 2: Regionalising Firm Emissions

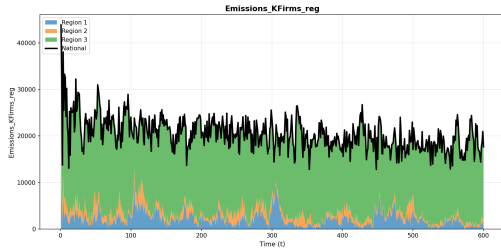


Figure: K firm Emissions

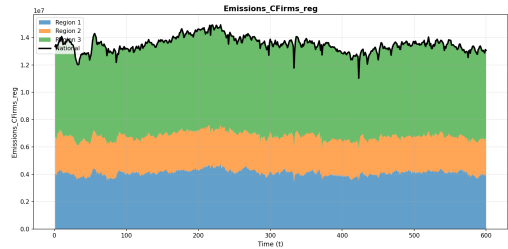


Figure: C firm Emissions

Sprint 2: Regionalising Emissions

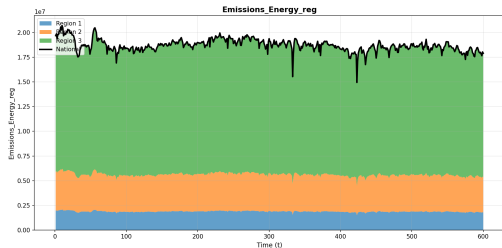


Figure: Energy Sector Emissions

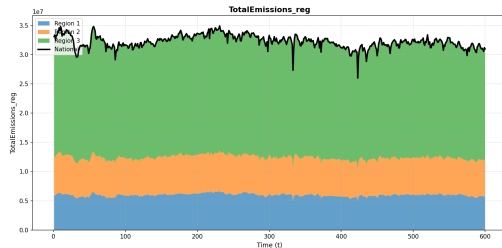


Figure: Total Emissions

Sprint 2: Regional Accounting of GDP

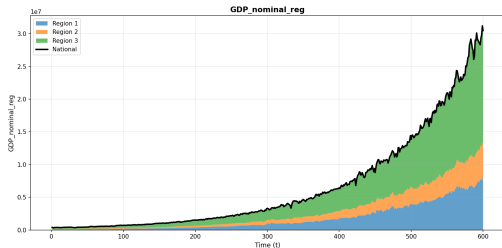


Figure: Nominal GDP

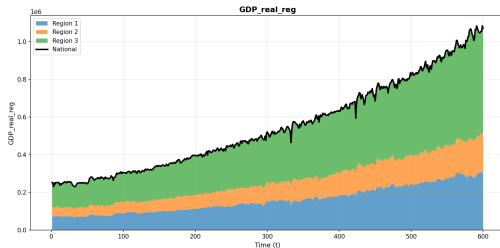


Figure: Real GDP

Regional Shocks

Sprint 3: Climate Shock Channels in Reissl et al. (2025)

- All firms in the region receive uniform shocks `flag_uniform_shock = 1`
 - 1 Shocks to productivity affecting the characteristics of capital vintages `flag_prodshocks1`
 - 2 Shocks to productivity (not affecting characteristics of capital vintages) `flag_prodshocks2`
 - 3 Energy Capacity both green and brown `flag_encapshocks`
 - 4 Shock to labour population `flag_popshocks`
 - 5 Shock to aggregate consumption demand `flag_demandshocks`
 - 6 Shock to C-Firms capital stock `flag_capshocks`
 - 7 Output shocks to C & K Firms `flag_outputshocks`
 - 8 Shock to C-Firms inventories `flag_inventshocks`
 - 9 Shock to R & D of K Firms `flag_RDshocks`

Sprint 3: Climate shocks in DSK

Climate shocks are modelled as stochastic shocks drawn from temperature-dependent Beta distributions in Reissl et al. (2025).

For each shock channel $s \in \{1, \dots, S\}$ and period t :

$$\varepsilon_{s,t} \sim \text{Beta}(a_{s,t}, b_{s,t}), \quad \varepsilon_{s,t} \in (0, 1)$$

The shape parameters evolve with global temperature T_t :

$$a_{s,t} = a_{0,s} \left[1 + \ln \left(\frac{T_{t-1}}{T_0} \right) \right]^{\theta_s^{(1)}}$$

$$b_{s,t} = b_{0,s} \left(\frac{T_0}{T_{t-1}} \right)^{\theta_s^{(2)}}$$

- $a_{0,s}, b_{0,s}$ are baseline shape parameters,
 - $a_{s,t}$ governs mean/median of damages,
 - $b_{s,t}$ governs right tail thickness (extreme events),
- $\theta_s^{(1)}$ controls increase in typical damages,
- $\theta_s^{(2)}$ controls amplification of extremes,
- T_t is global mean temperature, T_0 is reference level.
- Climate shocks affect real economic variables (e.g. X) multiplicatively:

$$X_{i,t} = X_{i,t-1} (1 - \varepsilon_{s,t})$$

Sprint 3: Parameters of Climate shock β distribution

- 1 **Global temperature** (T_t) is common to all regions
- 2 **Shock exponent** $\theta_s^{(1)}$ **and baseline severity parameters** $a_{0,s}$ have regional values but are kept same common across regions as they reflect physical damage sensitivities (how temperature translates into damage)
- 3 **Baseline exposure** ($b_{0,r}$): Region-specific right-tail thickness capturing structural exposure to extreme events (e.g. coastal vs inland regions).
- 4 **Extreme-event amplification** ($\theta_r^{(2)}$): Region-specific scaling of tail risk as temperature rises, allowing extreme damages to accelerate faster in some regions.

Sprint 3: Regionalisation of climate shock channels

- In the regional extension, climate shocks are generated **at the regional level** and propagated to all firms located in that region.
- For each shock channel $s \in \{1, \dots, S\}$, region r , and period t :

$$\varepsilon_{s,r,t} \sim \text{Beta}(a_{s,r,t}, b_{s,r,t})$$

- The realised regional shock $\varepsilon_{s,r,t}$ is then broadcast to all firms (or specified channel) i in region r :

$$X_{i,t} = X_{i,t-1}(1 - \varepsilon_{s,r,t}), \quad i \in r$$

- This mechanism is identical to Reissl et al. (2025).

Sprint 3: Regionalisation of climate shock channels

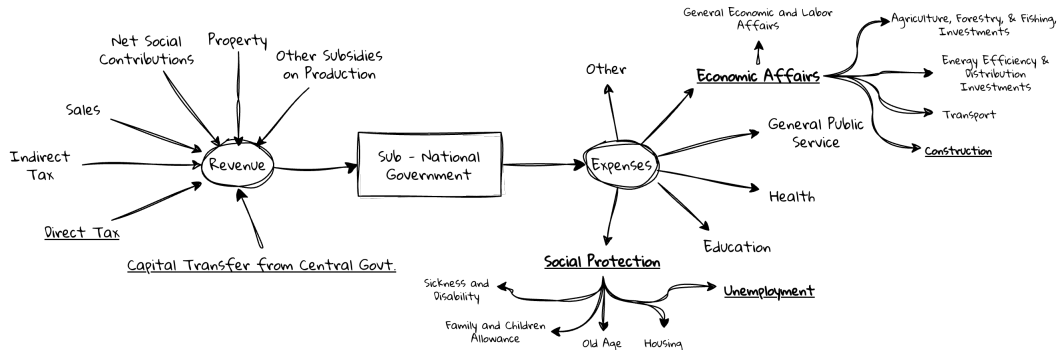
Regional Climate Shocks

- At each period, we draw one climate shock per region and per channel, driven by a common global temperature path and applied uniformly to all firms in the region.
- Regions differ in how strongly specific climate shock types hit (damages) and scale with warming, but all regions share the same global climate forcing.

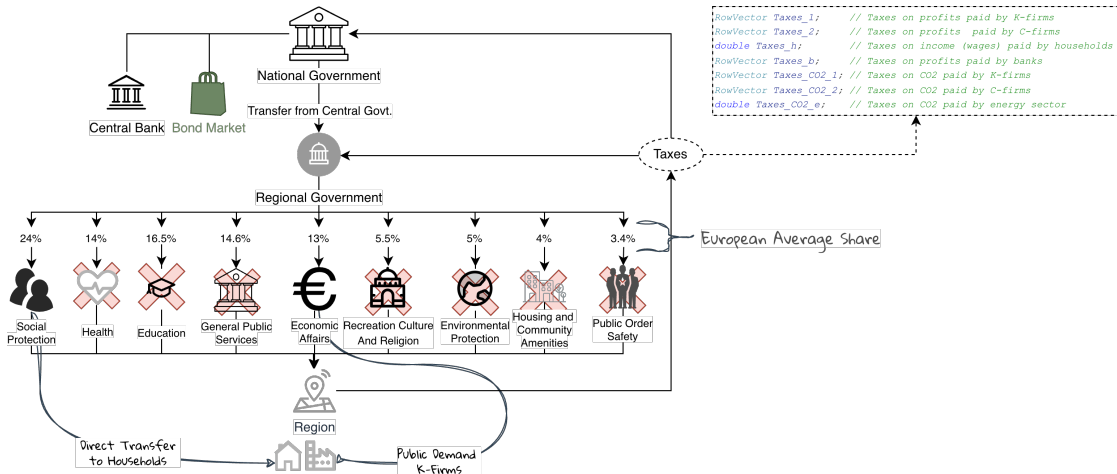
Regional Govt.

Conceptualising Sub-National Government in European Context

- Each country creates its own arrangement for dividing responsibilities between regional and local entities which can be collectively called as sub-national governments (Halaskova et al., 2021)
- Expenses and revenues of sub-national government can be calibrated using subnational government finances from ECB Olsson and Catz (2023)



Regional Govt. Revenue & Expenditure



Regional Government Revenue Design

$$TaxShare_{r,t} = \tau_r^{share} (T_{r,t}^{Cfirms} + T_{r,t}^{Kfims})$$

$$GRANTPOOL_t = \bar{\gamma} \cdot REVENUE_{CentralGovt,t}$$

$$GovtTransfer_{r,t}^{base} = \omega_r \cdot GRANTPOOL_t$$

$$GovtTransfer_{r,t}^{topup} = \max \left\{ 0, SocialProtection_{r,t} - (TaxShare_{r,t} + GovtTransfer_{r,t}^{base}) \right\}$$

$$REVENUE_{RegGovt,r,t} = GovtTransfer_{r,t}^{base} + GovtTransfer_{r,t}^{topup} + TaxShare_{r,t}$$

- $TaxShare_{r,t}$: Region's share of centrally collected taxes
- $GovtTransfer_{r,t}^{base}$: Standard intergovernmental transfer from grant pool
- $GovtTransfer_{r,t}^{topup}$: Additional transfer ensuring minimum welfare provision
- $REVENUE_{RegGovt,r,t}$: Total regional government revenue
- National government collects all taxes and redistributes to regions
- Grant pool bounded: $\bar{\gamma} \in [0, 1]$
- Allocation weights: $\sum_r \omega_r = 1$
- Welfare guarantee:
 $REV_{rg,r,t} \geq SP_{r,t}^{plan}$

Regional Government Expenditure Design

$$EXPENDITURE_{RegGovt,r,t} = SocialProtection_{r,t} + EconomicAffairs_{r,t}$$

Social Protection

$$SocialProtection_{r,t} = LS_{r,t} \cdot U_{r,t} \cdot w_{t-1} \cdot \omega_{u,r}$$

- $LS_{r,t}$: Regional labour supply pool
- $U_{r,t}$: Regional unemployment rate
- w_{t-1} : Wages
- $\omega_{u,r}$: Regional wage replacement rate

Economic Affairs

$$EconomicAffairs_{r,t} =$$

$$\max \{0, REVENUE_{r,t} - SocialProtection_{r,t}\}$$

- Public investment expenditure for building public stock
- For each K-firm in region r :

$$EconomicAffairs_{Kfirm,r,t} = \frac{EconomicAffairs_{r,t}}{N_{r,t}^{Kfirm}}$$

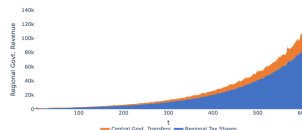
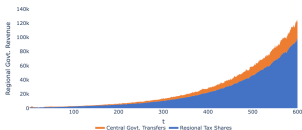
Regional Government Revenue and Expenditure Components

$$REVENUE_{RegGovt,r,t} = TaxShare_{r,t} + GovtTransfer_{r,t}$$

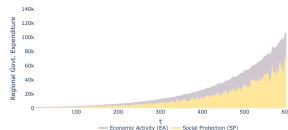
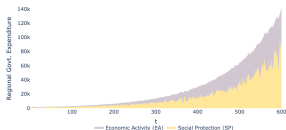
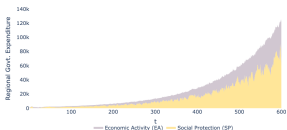
No Shock

CS

LP + EF



$$EXPENDITURE_{RegGovt,r,t} = SocialProtection_{r,t} + EconomicAffairs_{r,t}$$



$$REVENUE_{RegGovt,r,t} = EXPENDITURE_{RegGovt,r,t}$$

Above values refer to mean of Monte Carlo simulation runs of size 20.

Verification

Verification

To ensure model behaviours remains unchanged after introduction of regional government, we perform following simulation experiment. **Models** : (1) DSK (Reissl et al., 2025) (2) Regional - DSK

Setup

Scenarios

- ① No Shock
- ② Capital Shock
- ③ Labour Productivity + Energy Efficiency Shock

Unemployment Benefits routed through Regional Govt.

- DSK: $w_u = 0.4$
- Regional DSK: $w_u = 0$; $w_{u,r} = 0.4$

Regional Govt. Setup

- **Revenue**
 - $TaxShare_{r,t} = \tau_r^{share} (T_{r,t}^{Cfirms} + T_{r,t}^{Kfirms}) = 0$
 - Central Govt. Transfers = Regional Unemployment Benefits
- **Expense**
 - Social Protection = Unemployment benefits
 - Economic Affairs: We set regional tax share zero so that regional govt. relies entirely on central govt. transfers. This mandates that regional govt. never goes in surplus and thus has zero EA expense.

Verification | Macro Performance (R-DSK relative to DSK)

Metric	No Shock	CS	LP + EF
Unemployment	0.9980(0.21)	0.9934(0.79)	0.9961(0.38)
GDP Volatility	0.9932(0.45)	0.9976(0.16)	1.0094(− 0.65)
GDP Growth	0.9986(0.13)	1.0002(− 0.02)	1.0048(− 0.41)
Crisis Likelihood	0.9809(1.00)	0.9946(0.29)	1.0030(− 0.17)
Emissions	0.9906(0.42)	0.9672(1.56)	0.9866(0.62)

Entries report mean *Regional-DSK/DSK* for $n = 50$ monte-carlo simulations.

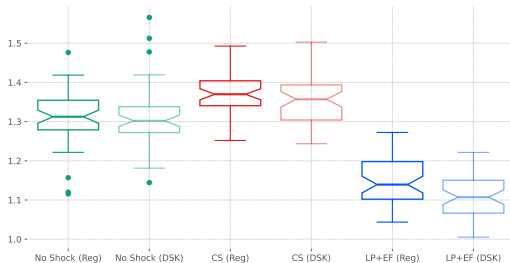
t -statistics for H_0 : equal means are in parentheses.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

	No Shock
$\frac{\text{RegGov SocialProtection (R-DSK)}}{\text{UnemploymentBenefit (DSK)}}$	0.9975
t -statistic	0.2036

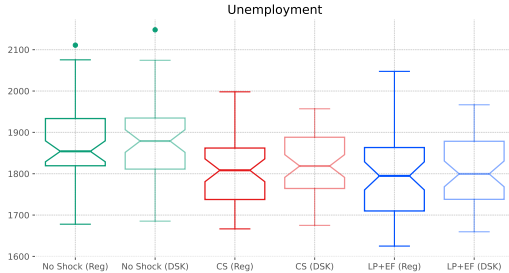
Verification | GDP Growth

Real GDP



Verification | Unemployment

Unemployment



Verification | Net Worth Comparison I

Net worth of Households

Net worth of C Firms

Scenario	Ratio $R\text{-}DSK/DSK$
No Shock	1.0016(− 0.13)
CS	0.9878(0.90)
LP + EF	1.0133(− 1.10)

Scenario	Ratio $R\text{-}DSK/DSK$
No Shock	1.0112(− 0.76)
CS	1.0158(− 1.01)
LP + EF	1.0281(− 1.93)

Welch's t -test ($n = 50$). *** $p < 0.01$.

Verification | Net Worth Comparison II

Net worth of K Firms

Net worth of *CentralGovt./GDP*

Scenario	Ratio $R\text{-}DSK/DSK$
No Shock	0.9667(1.05)
CS	1.0074(− 0.22)
LP + EF	0.9808(0.59)

Scenario	Ratio $R\text{-}DSK/DSK$
No Shock	0.9909(− 0.66)
CS	0.9759(− 1.56)
LP + EF	0.9830(− 1.43)

Welch's t -test ($n = 50$). *** $p < 0.01$.

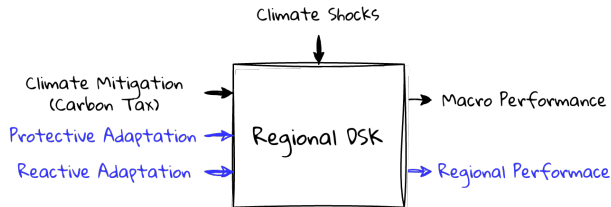
Adaptation

Conceptualising Climate Adaptation Module in R-DSK

- DSK (Reissl et al., 2025) has climate shock propagation and mitigation policy levers but no explicit climate adaptation
- **Objective**
 - Integrate explicit, regional, and dynamic adaptation
 - Preserve stochastic micro-damage structural approach of climate shock function
 - Ensure stock–flow consistency

- **Approach**

- Adaptation modifies:
 - Exposure (through protection)
 - Recovery dynamics (re-building)



Climate Adaptation: Conceptual Approaches in IAMs

1. Adaptation as residual (implicit)

- Embedded in aggregate damage functions (net damages)
- No explicit modelling of behaviour, policy, or capacity; cannot analyse adaptation decisions or heterogeneity Patt et al. (2010)

2. Adaptation as an optimisation decision

- Trade-off between adaptation investment and residual damages
- Includes *flow* (reactive) and *stock* (capacity-building) adaptation, but remains highly aggregated and representative-agent based (Bruin et al., 2007)

3. Adaptation as sectoral and spatially heterogeneous

- Implemented in specific sectors or regions (e.g. agriculture, coastal, urban systems)
- Recognises the local and context-specific nature of adaptation processes Füssel (2010); Meng et al. (2020)

4. Adaptation as dynamic capacity and coping range

- Damages depend on climate relative to a time-varying coping range
- Distinguishes *exposure* from *adaptive capacity*; captures acclimatisation and resilience dynamics (Estrada et al., 2019)

5. Adaptation as constrained by capacity, institutions, and barriers

- Adaptation depends on finance, governance, knowledge, and implementation capacity
- Barriers prevent adaptation even when it is welfare-improving Eisenack et al. (2014); Andrijevic et al. (2023)

6. Adaptation as limited and potentially maladaptive

- Adaptation can be path-dependent, uneven, and subject to hard or soft limits
- Some responses may reduce risk in the short run while increasing long-run vulnerability Berkhout and Dow (2023); Reckien et al. (2023)

Regional Adaptation Spending Rule

- Adaptation is modelled as a combination of stock (planned capacity-building) and flow (reactive) expenditures, where investment in adaptation can contribute to the accumulation of resilience over time Bruin et al. (2007); Füssel (2010); Monsod et al. (2023)
- Public spending increases after disasters (“build back better”) (Hochrainer-Stigler et al., 2020; Bayar and Yarbrough, 2024)
- Adaptation capacity strongly linked to income and fiscal space (Saeed et al., 2023)

Regional Government Adaptation Spending

Regional Government Expenditure

$$EXP_{r,t} = SP_{r,t} + I_{r,t}^{Adapt} + EA_{r,t}^{res}$$

$$EA_{RegGovt,r,t}^{res} = \max\{0, REV_{r,t} - SP_{r,t} - I_{r,t}^{Adapt}\}$$

$EA_{r,t}^{res}$ is residual productive expenditure after adaptation commitments are met.

Adaptation investment

$$I_{r,t}^{Adapt} = I_{r,t}^{Prot} + I_{r,t}^{Rec}$$

$$I_{r,t}^{Prot} = \iota_r^A \cdot Y_{r,t-1}$$

$$I_{r,t}^{Rec} = \psi_r^A \cdot Y_{r,t-1} \cdot \max(0, \tilde{S}_{r,t-1} - \bar{s}_r)$$

Central Government Funding Post-Disaster Recovery

If fiscal space is insufficient to fund $I_{r,t}^{Rec}$, the central government extends a recovery top-up from the national grant pool ($GRANTPOOL_t = \bar{\gamma} \mathcal{T}_t$):

$$GT_{r,t}^{Rec} = \max \left(0, I_{r,t}^{Rec} - \max \left(0, REV_{r,t} - SP_{r,t} \right) \right)$$

This prevents the perverse feedback in which large shocks simultaneously destroy regional tax revenues **and** make recovery spending infeasible.

Grant pool priority ordering

$$\underbrace{GT_{r,t}^{SP}}_{\text{SP top-up (existing)}} \succ \underbrace{GT_{r,t}^{Rec}}_{\text{Recovery top-up (new)}} \succ \underbrace{GT_{r,t}^{base}}_{\text{Base grant for Protective Adaptation / EA}}$$

Both SP and recovery are guaranteed obligations of the centre; planned protection is funded from the residual base allocation.

Protection: Planned Adaptation

$$I_{r,t}^{Prot} = \iota_r^A \cdot Y_{r,t-1}$$

- ι_r^A : exogenous region-specific planned protection intensity (share of lagged real output); scaled so that larger/wealthier regions sustain proportionally larger protection budgets
- Implemented as a government order placed with K-firms in the region: the government credits K-firm deposit accounts by $I_{r,t}^{Prot} / N_{1,r,t}$ per firm, generating demand for physical protection capital (flood defences, drainage, resilient infrastructure) in the capital-goods sector

Protection (adaptation) capital accumulation

$$K_{r,t}^{Prot} = (1 - \delta_{Prot}) K_{r,t-1}^{Prot} \cdot I_{r,t}^{Prot}$$

Protection: Climate Shocks under Adaptation

Exposure multiplier

$$\Omega_{r,t} = \bar{\omega}_r + (1 - \bar{\omega}_r) \exp \left(-\phi_P \frac{K_{r,t}^{Prot}}{Y_{r,t-1}} \right) \in (\bar{\omega}_r, 1]$$

- $\Omega_{r,t} = 1$: no accumulated protection ($K^{Prot} = 0$) — full exposure
- $\Omega_{r,t} \rightarrow \bar{\omega}_r > 0$: maximum protection stock — **residual exposure always remains**; even perfect infrastructure does not eliminate stochastic losses
- $\phi_P > 0$: efficiency of protection capital at reducing exposure (calibrated parameter)

Effect on realised shocks

$$shock_{r,j,t}^{eff} = \Omega_{r,t} \cdot shock_{r,j,t}^{raw}$$

$$shock_{r,j,t}^{raw} \sim \text{Beta}(Xa_{r,t}^s, Xb_{r,t}^s)$$

Protection compresses the shock realisation by $\Omega_{r,t} < 1$, but since $shock^{raw}$ is drawn from a Beta distribution, large realisations still pass through. Thus, protection is probabilistic, not deterministic.

Recovery: Reactive Post-Shock Adaptation

$$I_{r,t}^{Rec} = \psi_r^A \cdot Y_{r,t-1} \cdot \max(0, \tilde{S}_{r,t-1} - \bar{s}_r)$$

- ψ_r^A : exogenous region-specific recovery responsiveness
- $\tilde{S}_{r,t-1}$: realised post-adaptation regional capital shock index in the previous period (see below)
- $\bar{s}_r$: minimum shock threshold below which no reconstruction programme is initiated

Recovery mechanism: C-firm investment subsidy for machine purchases

Recovery is modelled as a **per-firm cash transfer to C-firms** in the affected region, supplementing their machine-purchase budget for the following period

$$sub_{j,t}^{Rec} = \frac{I_{r,t}^{Rec}}{N_{2,r,t-1}}, \quad j \in r$$

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